
MIDUS Refresher 2 Cognitive Project

Variable Naming Cognitive Test Battery

Brief Test of Adult Cognition by Telephone (BTACT) Montreal Cognitive Assessment (MoCA-BLIND)

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Brief Test of Adult Cognition by Telephone (BTACT)

A. Coding conventions for BTACT data

- a) YES = 1, NO = 2
- b) PROBLEM HEARING/UNDERSTANDING/RECORDING: 96
- c) DON'T KNOW (7's): 7, 97, 997, 9997
- d) REFUSED OR MISSING (8's): 8, 98, 998, 9998

B. Naming conventions for BTACT variables

For the MIDUS Refresher 2 the first two characters of each variable name will be "RB". Otherwise, the same naming conventions developed for MIDUS 2 apply.

- a) 1st letter – "R," to indicate MIDUS Refresher
- b) 2nd letter – "B" to indicate Wave 2
- b) 3rd letter – indicates Project Number 3
- c) 4th letter – indicates type of test: T= BTACT cognitive tests
- d) 5th, 6th, 7th, 8th, 9th letters – indicates either:
 - i. The individual test:
 - WLI: Word List Recall – Immediate (Note: Columns 7 & 8 indicate response number; 1-26 possible responses)
 - BD: Backward Digit Span
 - CTFL: Category Fluency (also CTF)
 - NS: Number Series (Note: Column 6 indicates trial number 1-5)
 - BK: Backward Counting
 - WLD: Word List Recall – Delayed (Note: Columns 7 & 8 indicate response number; 1-26 possible responses)
 - ii. A composite measure:
 - WLF: Word List: Proportion Forgotten Between Immediate and Delayed
 - COMP: BTACT Composite Score
 - iii. A flag variable: Variables with "FP" as their last characters serve as flag variables for potentially problematic cases. This variable indicates, by test, which cases were identified at Brandeis by our data cleaning as being problematic due to test disruption, interview equipment failures, or other problems. We recommend users exclude these tests for these specific cases.

Note: **Bold** variable names below indicate composite or total scale scores.

VARIABLE NAME	VARIABLE LABEL	VALUES
Word List Recall: Immediate		
RB3TWLIFP	Word List Immediate flagged problematic?	1=YES 2=NO 8=REFUSED OR MISSING
RB3TWLI1 ... RB3TWLI26	Word List Immediate: Recalled #1 ... #26 (allows for max 15 correct responses + up to 11 intrusions/repetitions)	1=WORD 1 2=WORD 2 3=WORD 3 4=WORD 4 5=WORD 5 6=WORD 6 7=WORD 7 8=WORD 8 9=WORD 9 10=WORD 10 11=WORD 11 12=WORD 12 13=WORD 13 14=WORD 14 15=WORD 15 90=NON-LIST INTRUSION 96=PROBLEM HEARING/ UNDERSTANDING/RECORDING 98=REFUSED OR MISSING
RB3TWLITU	Word List Immediate: Total Unique Items	Range: 0 to 15; Sum of all correct, unique responses from RB3TWLI1 to RB3TWLI26 98 = REFUSED OR MISSING
RB3TWLITR	Word List Immediate: Total repetitions	Sum of all repeated responses from RB3TWLI1 to RB3TWLI26 98 = REFUSED OR MISSING
RB3TWLITI	Word List Immediate: Total intrusions	Sum of all non-list intrusions from RB3TWLI1 to RB3TWLI26 98 = REFUSED OR MISSING
Digits Backward		
RB3TDBFP	Digits Backward flagged problematic?	1=YES 2=NO 8=REFUSED OR MISSING
B3TDBS	Digits Backward: highest # digits recall	1 to 8 98 = REFUSED OR MISSING

Category Fluency		
RB3TCTFFP	Category Fluency flagged problematic?	1=YES 2=NO 8=REFUSED OR MISSING
RB3TCTFLU	Category Fluency: Tot Unique Items	Sum of all in-category, unique items named 98 = REFUSED OR MISSING
RB3TCTFLR	Category Fluency: Total repetitions	Sum of repeated items named in-category 98 = REFUSED OR MISSING
RB3TCTFLI	Category Fluency: Total intrusions	Sum of all non-category intrusions 98 = REFUSED OR MISSING
Number Series		
RB3TNSFP	Number Series flagged problematic?	1=YES 2=NO 8=REFUSED OR MISSING
RB3TNS1 ... RB3TNS5	Number Series 1...5 (number reported)	998 = REFUSED OR MISSING
RB3TNS1C... RB3TNS5C	Number Series 1...5: correct?	1=YES, CORRECT 2=NO, INCORRECT 8=REFUSED OR MISSING
RB3TNSTOT	Number Series: Total Correct	Range: 0 to 5; 8=REFUSED OR MISSING
Backward Counting		
RB3TBKFP	Backward Counting flagged problematic?	1=YES 2=NO 8=MISSING
RB3TBKCT	Backward Counting: last number reached	998 = REFUSED OR MISSING
RB3TBKERR	Backward Counting: number of errors	998 = REFUSED OR MISSING
RB3TBKTOT	Backward counting: total number counted correctly: (100-(RB3TBKCT + RB3TBKERR))	Total correct #s produced 998 = REFUSED OR MISSING
Word List Recall-Delayed		
RB3TWLDFP	Word List Delayed flagged problematic?	1=YES 2=NO 8=REFUSED OR MISSING

RB3TWLD1... RB3TWLD26	Word List Delayed: Recalled #1...#26 (allows for max 15 correct responses + up to 11 intrusions/repetitions)	1=WORD 1 2=WORD 2 3=WORD 3 4=WORD 4 5=WORD 5 6=WORD 6 7=WORD 7 8=WORD 8 9=WORD 9 10=WORD 10 11=WORD 11 12=WORD 12 13=WORD 13 14=WORD 14 15=WORD 15 90=NON-LIST INTRUSION 96=PROBLEM HEARING/UNDERSTAND/ RECORDING 98 = REFUSED OR MISSING
RB3TWLDTU	Word List Delayed: Total unique items	Range: 0 to 15; Sum of all correct, unique responses from RB3TWLD1 to RB3TWLD26 98 = REFUSED OR MISSING
RB3TWLDTR	Word List Delayed: Total repetitions	Sum of all repeated responses from RB3TWLD1 to RB3TWLD26 98 = REFUSED OR MISSING
RB3TWLDTI	Word List Delayed: Total intrusions	Sum of all non-list intrusions from RB3TWLD1 to RB3TWLD26 98 = REFUSED OR MISSING
Composite Measures		
RB3TWLFF	Word List: Proportion Forgotten Between Immediate and Delayed	$(RB3TWLITU - RB3TWLDTU) / RB3TWLITU$ 8=REFUSED OR MISSING
RB3TCOMPZ_Long	Longitudinal BTACT Composite Score (standardized based on R1 means and SDs)	Standardized mean of z-scores* based on R1 values for Word Lists (sum of Immediate and Delayed: RB3TWLITU + RB3TWLDTU), Digits Backward (RB3TDBS), Category Fluency (RB3TCTFLU), Number Series (RB3TNSTOT), and Backward Counting (RB3TBKTOT) 8=REFUSED OR MISSING

RB3TEMZ_Long	Longitudinal BTACT Episodic Memory Factor (standardized based on R1 means and SDs)	Standardized mean of z-scores* based on R1 values for Word List Immediate (RB3TWLITU) and Word List Delayed (RB3TWLDTU) 8=REFUSED OR MISSING
RB3TEFZC_Long	Longitudinal BTACT Executive Functioning Factor (standardized based on R1 means and SDs) corrected based on the metronome values	Standardized mean of z-scores* based on R1 values for Digits Backward (RB3TDBS), Category Fluency (RB3TCTFLU), Number Series (RB3TNSTOT), Backward Counting (RB3TBKTOT), and mean of switch and nonswitch trials (RB3TSMXBBC multiplied by -1 and z-scored) in the Stop & Go Switch Task (SGST)† 8=REFUSED OR MISSING

*The z-scores were computed using the means and standard deviations obtained on the main Refresher 1 sample (N = 2550). Please see Data File Notes for longitudinal restandardization information.

† See description of SGST variables below. Also note the recommendation to use the RB3TSFC filter described at the end of this document when working with SGST variables.

STOP AND GO SWITCH TASK

A. Coding convention for SGST data

a) **INVALID OR MISSING (8's): 8, 98, 998**

B. Naming conventions for Stop and Go Switch Task (SGST) variables

a) Naming convention for individual trials (raw scores)

- i. 1st, 2nd, 3rd, and 4th character: by default is the **RB3T** code that indicates MIDUS Refresher, Wave 2, Project 3, BTACT Cognitive Battery
 - R = MIDUS Refresher
 - B = Wave 2
 - 3 = Project 3
 - T = BTACT Cognitive Battery
- ii. 5th character indicates the **Stop & Go Switch Task**
- iii. 6th character: W = Ra**W** scores
- iv. 7th character indicates the subtest
 - N = **N**ormal single-task
 - R = **R**everse single-task
 - X = **miX**ed-task
- v. 8th and 9th character indicate trial number (**1-20** for single-task, **1-32** for mixed-task)

b) Naming convention for Normal and Reverse single-tasks composite scores

- i. 1st, 2nd, and 3rd, and 4th character, by default, is the **RB3T** code that indicates MIDUS Refresher, Wave 2, Project 3, BTACT Cognitive Battery
 - R = MIDUS Refresher
 - B = Wave 2
 - 3 = Project 3
 - T = BTACT Cognitive Battery
- ii. 5th character indicates the **Stop & Go Switch Task**
- iii. 6th character indicates measure
 1. For accuracy scores
 - T = **T**otal correct
 - V = **i**nvalid
 - P = **P**ercent correct
 2. For latency scores
 - M = **M**edian (or mean of medians)
- iv. 7th character indicates the subtest
 - N = **N**ormal single-task
 - R = **R**everse single-task
- v. 8th character indicates the scores corrected based on the metronome

values

C = **C**orrected

c) Naming convention for Mixed-task composite scores

- i. 1st, 2nd, 3rd, and 4th character, by default, is the **RB3T** code that indicates MIDUS 3, Project 3, BTACT Cognitive Battery
 - R = MIDUS Refresher
 - B = Wave 2
 - 3 = Project 3
 - T = BTACT Cognitive Battery
- ii. 5th character indicates the **S**top & Go Switch Task
- iii. 6th character indicates measure
 1. For accuracy scores
 - T = **T**otal correct
 - V = **i**nvalid
 - P = **P**ercent correct
 2. For latency scores
 - M = **M**edian (or mean of medians)
- iv. 7th character indicates the subtest
 - X = **m**iXed-task
- v. 8th character indicates the condition
 - N = **N**ormal
 - R = **R**everse
 - B = combined
- vi. 9th character indicates the trial type
 - S = **S**witch
 - O = **n**Oswitch
 - B = combined
- vii. 10th character indicates the scores corrected based on the metronome values
 - C = **C**orrected

d) Naming convention for metronome tasks

- i. 1st, 2nd, 3rd, and 4th character, by default, is the **RB3T** code that indicates MIDUS Refresher, Wave 2, Project 3, BTACT Cognitive Battery
 - R = MIDUS Refresher
 - B = Wave 2
 - 3 = Project 3
 - T = BTACT Cognitive Battery
- ii. 5th character indicates the **S**top & Go Switch Task
- iii. 6th character indicates measure
 - M = **M**edian
- iv. 7th character indicates the subtest
 - M = **M**etronome

e) Composite scores: variable names

i. Accuracy

1. **RB3TSPN**: normal single-task % correct
2. **RB3TSPR**: reverse single-task % correct
3. **RB3TSPXNO**: mixed-task normal nonswitch % correct
4. **RB3TSPXRO**: mixed-task reverse nonswitch % correct
5. **RB3TSPXBO**: mixed-task nonswitch % correct
6. **RB3TSPXNS**: mixed-task normal switch % correct
7. **RB3TSPXRS**: mixed-task reverse switch % correct
8. **RB3TSPXBS**: mixed-task switch % correct
9. **RB3TSPXBB**: all mixed-task trials % correct

ii. Latencies

1. **RB3TSMN**: normal single-task median RT (reaction time)
2. **RB3TSMR**: reverse single-task median RT (reaction time)
3. **RB3TSMB**: mean(RB3TSMN, RB3TSMR) (normal and reverse)
4. **RB3TSMXNO**: mixed-task normal nonswitch median RT
5. **RB3TSMXRO**: mixed-task reverse nonswitch median RT
6. **RB3TSMXBO**: median RT of all mixed-task nonswitch
7. **RB3TSMXNS**: mixed-task normal switch median RT
8. **RB3TSMXRS**: mixed-task reverse switch median RT
9. **RB3TSMXBS**: median RT of all mixed-task switch
10. **RB3TSMXBB**: mean(RB3TSMXBO, RB3TSMXBS) (nonswitch and switch trials)

iii. Latencies corrected based on the metronome values

1. **RB3TSMNC** = RB3TSMN - RB3TSMNM.
2. **RB3TSMRC** = RB3TSMR - RB3TSMNM.
3. **RB3TSMBC** = mean(RB3TSMNC, RB3TSMRC) (normal and reverse)
4. **RB3TSMXNOC** = RB3TSMXNO - RB3TSMNM.
5. **RB3TSMXROC** = RB3TSMXRO - RB3TSMNM.
6. **RB3TSMXBOC** = RB3TSMXBO - RB3TSMNM.
7. **RB3TSMXNSC** = RB3TSMXNS - RB3TSMNM.
8. **RB3TSMXRSC** = RB3TSMXRS - RB3TSMNM.
9. **RB3TSMXBSC** = RB3TSMXBS - RB3TSMNM.
10. **RB3TSMXBBC** = mean(RB3TSMXBOC, RB3TSMXBSC) (nonswitch and switch trials)

f) Naming convention for cost variables

- i. 1st, 2nd, 3rd, 4th, and 5th characters: as above: **RB3TS**, for MIDUS Refresher, Wave 2, Project 3 BTACT Cognitive Battery, Stop & Go Switch Task
- ii. 6th character: C=Cost
- iii. 7th character: indicates G=General, L=Local
- iv. 8th character indicates the condition
N = Normal

R = Reverse

B = comBined

- v. 9th character: A = **A**bsolute cost, R = **R**elative cost
- vi. 10th character: C = **C**orrected based on the metronome values

g) Switch Costs: variable names

- i. General switch costs compare latencies on mixed-task trials to single-task trials (mean of normal single-task and reverse single-task). Although there are several ways of calculating general switch costs, we selected this version as the most basic. We give both *absolute* and *relative* general switch costs. *Absolute costs* represent a simple difference score between the easier and more difficult condition (e.g. A – B). *Relative costs* give the proportion decline in performance from the easier to the harder condition, and thus control for differences in baseline performance (e.g. (A-B)/A.)
 - 1. **RB3TSCGNA**: General Switch Cost (normal), absolute
[mean(RB3TSMXNO, RB3TSMXNS) – RB3TSMN]
 - 2. **RB3TSCGNR**: General Switch Cost (normal), relative
(RB3TSCGNA/RB3TSMN)
 - 3. **RB3TSCGRA**: General Switch Cost (reverse), absolute
[mean(RB3TSMXRO, RB3TSMXRS) – RB3TSMR]
 - 4. **RB3TSCGRR**: General Switch Cost (reverse), relative
(RB3TSCGRA/RB3TSMR)
 - 5. **RB3TSCGBA**: General Switch Cost (combined), absolute
(RB3TSMXBB-RB3TSMB)
 - 6. **RB3TSCGBR**: General Switch Cost (combined), relative
(RB3TSCGBA/RB3TSMB)
- ii. General switch costs corrected based on the metronome values
 - 1. **RB3TSCGNAC**: General Switch Cost (normal), absolute
[mean(RB3TSMXNOC, RB3TSMXNSC) – RB3TSMNC]
 - 2. **RB3TSCGNRC**: General Switch Cost (normal), relative
(RB3TSCGNAC/RB3TSMNC)
 - 3. **RB3TSCGRAC**: General Switch Cost (reverse), absolute
[mean(RB3TSMXROC, RB3TSMXRSC) – RB3TSMRC]
 - 4. **RB3TSCGRRC**: General Switch Cost (reverse), relative
(RB3TSCGRAC/RB3TSMRC)
 - 5. **RB3TSCGBAC**: General Switch Cost (combined), absolute
(RB3TSMXBBC-RB3TSMBC)
 - 6. **RB3TSCGBRC**: General Switch Cost (combined), relative
(RB3TSCGBAC/RB3TSMBC)
- iii. Local switch costs compare mixed-task switch trials to mixed-task nonswitch trials. We give both *absolute* local switch costs and *relative* local switch costs.

1. **RB3TSCLNA**: Local Switch Cost (normal), absolute
(RB3TSMXNS – RB3TSMXNO)
2. **RB3TSCLNR**: Local Switch Cost (normal), relative
(RB3TSCLNA/RB3TSMXNO)
3. **RB3TSCLRA**: Local Switch Cost (reverse), absolute
(RB3TSMXRS – RB3TSMXRO)
4. **RB3TSCLRR**: Local Switch Cost (reverse), relative
(RB3TSCLRA/RB3TSMXRO)
5. **RB3TSCLBA**: Local Switch Cost (combined), absolute
(RB3TSMXBS- RB3TSMXBO)
6. **RB3TSCLBR**: Local Switch Cost (combined), relative
(RB3TSCLBA/RB3TSMXBO)

iv. Local switch costs corrected based on the metronome values

1. **RB3TSCLNAC**: Local Switch Cost (normal), absolute
(RB3TSMXNSC – RB3TSMXNOC)
2. **RB3TSCLNRC**: Local Switch Cost (normal), relative
(RB3TSCLNAC/RB3TSMXNOC)
3. **RB3TSCLRAC**: Local Switch Cost (reverse), absolute
(RB3TSMXRSC – RB3TSMXROC)
4. **RB3TSCLRRC**: Local Switch Cost (reverse), relative
(RB3TSCLRAC/RB3TSMXROC)
5. **RB3TSCLBAC**: Local Switch Cost (combined), absolute
(RB3TSMXBSC- RB3TSMXBOC)
6. **RB3TSCLBRC**: Local Switch Cost (combined), relative
(RB3TSCLBAC/RB3TSMXBOC)

h) **Filters**: We provide two levels of filters. Researchers who wish to use all valid files can choose to select the Valid filter (RB3TSFV below). In our analyses, we have used a criterion of 75% accuracy to ensure that participants were performing the task correctly; researchers who wish to use this approach can select cases based on the Clean filter (RB3TSFC below).

- i. RB3TSFV (**Valid**): filters cases that were invalid due to missing sound files, technical problems, or failure to carry out the task as instructed.
- ii. RB3TSFC (**Clean**): To further insure that participants were performing the task as directed, we required a valid file with accuracy of at least 75% on all conditions (normal single-task, reverse single-task, mixed-task switch and nonswitch). In addition, to eliminate extreme latencies (i.e., outliers), we required median values of <2 sec for single-task and <4 sec for mixed-task trials.

Note: **Bold** variable names below indicate composite or total scale scores.

VARIABLE NAME	VARIABLE LABEL	VALUES
Individual Trials: Raw Scores		
RB3TSWN1... RB3TSWN20	SGST: normal single-task trial #1...#20	Latencies (s) in seconds 98 = INVALID OR MISSING
RB3TSWR1... RB3TSWR20	SGST: reverse single-task trial #1...#20	Latencies (s) in seconds 98 = INVALID OR MISSING
RB3TSWX1... RB3TSWX32	SGST: mixed-task trial #1 “normal...green” ...trial #32 “green”	Latencies (s) in seconds 98 = INVALID OR MISSING
Normal Single-task Trials: Composite Scores		
Composite Accuracy Scores		
RB3TSTN	SGST: normal single-task number correct	0-20 98 = REFUSED OR MISSING
RB3TSVN	SGST: normal single-task number invalid	0-20 98 = REFUSED OR MISSING
RB3TSPN	SGST: normal single-task percent correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Composite Latency Score		
RB3TSMN	SGST: normal single-task median RT (reaction time)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMNC	SGST: normal single-task median RT (reaction time) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Reverse Single-task Trials: Composite Scores		
Composite Accuracy Scores		
RB3TSTR	SGST: reverse single-task number correct	0-20 98 = REFUSED OR MISSING
RB3TSVR	SGST: reverse single-task number invalid	0-20 98 = REFUSED OR MISSING
RB3TSPR	SGST: reverse single-task percent correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Composite Latency Score		
RB3TSMR	SGST: reverse single-task median RT (reaction time)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMRC	SGST: reverse single-task median RT (reaction time) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Composite of Normal and Reverse Single-task		
RB3TSMB	SGST: mean(RB3TSMN, RB3TSMR)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMBC	SGST: mean(RB3TSMNC, RB3TSMRC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING

Mixed-task Trials: Composite Accuracy Scores		
Accuracy composite across normal nonswitch trials		
RB3TSTXNO	SGST: mixed-task normal nonswitch trials number correct	0-12 98 = REFUSED OR MISSING
RB3TSVXNO	SGST: mixed-task normal nonswitch trials number invalid	0-12 98 = REFUSED OR MISSING
RB3TSPXNO	SGST: mixed-task normal nonswitch trials percent correct (ratio form)	0-1.00 8 = INVALID OR MISSING
Accuracy composite across reverse nonswitch trials		
RB3TSTXRO	SGST: mixed-task reverse nonswitch trials number correct	0-11 98 = REFUSED OR MISSING
RB3TSVXRO	SGST: mixed-task reverse nonswitch trials number invalid	0-11 98 = REFUSED OR MISSING
RB3TSPXRO	SGST: mixed-task reverse nonswitch trials percent correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Accuracy composites across all nonswitch trials		
RB3TSTXBO	SGST: mixed-task nonswitch trials number correct	0-23 98 = REFUSED OR MISSING
RB3TSVXBO	SGST: mixed-task nonswitch trials number invalid	0-23 98 = REFUSED OR MISSING
RB3TSPXBO	SGST: mixed-task nonswitch trials percent correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Accuracy composite across normal switch trials		
RB3TSTXNS	SGST: mixed-task normal switch trials number correct	0-3 8 = REFUSED OR MISSING
RB3TSVXNS	SGST: mixed-task normal switch trials number invalid	0-3 8 = REFUSED OR MISSING
RB3TSPXNS	SGST: mixed-task normal switch trials percent correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Accuracy composite across reverse switch trials		
RB3TSTXRS	SGST: mixed-task reverse switch trials number correct	0-3 8 = REFUSED OR MISSING
RB3TSVXRS	SGST: mixed-task reverse switch trials number invalid	0-3 8 = REFUSED OR MISSING
RB3TSPXRS	SGST: mixed-task reverse switch trials percent correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Accuracy composite across all switch trials		
RB3TSTXBS	SGST: mixed-task switch trials number correct	0-6 8 = REFUSED OR MISSING
RB3TSVXBS	SGST: mixed-task switch trials number invalid	0-6 8 = REFUSED OR MISSING

RB3TSPXBS	SGST: mixed-task switch trials %correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Accuracy composites across all mixed-task trials		
RB3TSTXBB	SGST: all mixed-task number correct	0-29 98 = REFUSED OR MISSING
RB3TSVXBB	SGST: all mixed-task number invalid	0-29 98 = REFUSED OR MISSING
RB3TSPXBB	SGST: all mixed-task percent correct (ratio form)	0-1.00 8 = REFUSED OR MISSING
Mixed-task Trials: Composite Latency Scores		
Latency composite across normal nonswitch trials		
RB3TSMXNO	SGST: mixed-task normal nonswitch median RT	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMXNOC	SGST: mixed-task normal nonswitch median RT corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Latency composite across reverse nonswitch trials		
RB3TSMXRO	SGST: mixed-task reverse nonswitch median RT	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMXROC	SGST: mixed-task reverse nonswitch median RT corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Latency composite across all nonswitch trials		
RB3TSMXBO	SGST: mixed-task nonswitch trials median RT	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMXBOC	SGST: mixed-task nonswitch trials median RT corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Latency composite across normal switch trials		
RB3TSMXNS	SGST: mixed-task normal switch median RT	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMXNSC	SGST: mixed-task normal switch median RT corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Latency composite across reverse switch trials		
RB3TSMXRS	SGST: mixed-task reverse switch median RT	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMXRSC	SGST: mixed-task reverse switch median RT corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Latency composite across all switch trials		
RB3TSMXBS	SGST: mixed-task switch trials median RT	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSMXBSC	SGST: mixed-task switch trials median RT corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Latency composite across all mixed-task trials		
RB3TSMXBB	SGST: mean(B3TSMXBO, B3TSMXBS)	Latency (s) in seconds 98 = REFUSED OR MISSING

RB3TSMXBBC	SGST: mean(B3TSMXBOC, B3TSMXBSC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Mixed-Task Trials: Switch Cost Scores		
RB3TSCGBA	SGST: General Switch Cost, absolute (RB3TSMXBB-RB3TSMB)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGBR	SGST: General Switch Cost, relative (RB3TSCGBA/RB3TSMB)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGNA	SGST: General Switch Cost (normal), absolute [mean(RB3TSMXNO, RB3TSMXNS) – RB3TSMN]	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGNR	SGST: General Switch Cost (normal), relative (RB3TSCGNA/RB3TSMN)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGRA	SGST: General Switch Cost (reverse), absolute [mean(RB3TSMXRO, RB3TSMXRS) – RB3TSMR]	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGRR	SGST: General Switch Cost (reverse), relative (RB3TSCGRA/ RB3TSMR)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLBA	SGST: Local Switch Cost, absolute (RB3TSMXBS-RB3TSMXBO)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLBR	SGST: Local Switch Cost, relative (RB3TSCLBA/RB3TSMXBO)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLNA	SGST: Local Switch Cost (normal), absolute (RB3TSMXNS-RB3TSMXNO)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLNRR	SGST: Local Switch Cost, (normal), relative (RB3TSCLNA/RB3TSMXNO)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLRA	SGST: Local Switch Cost (reverse), absolute (RB3TSMXRS-RB3TSMXRO)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLRR	SGST: Local Switch Cost,(reverse), relative (RB3TSCLRA/RB3TSMXRO)	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGBAC	SGST: General Switch Cost, absolute (RB3TSMXBBC-RB3TSMBC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGBRC	SGST: General Switch Cost, relative (RB3TSCGBAC/RB3TSMBC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGNAC	SGST: General Switch Cost (normal), absolute [mean(RB3TSMXNOC, RB3TSMXNSC) – RB3TSMNC] corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGNRC	SGST: General Switch Cost (normal), relative (RB3TSCGNAC/RB3TSMNC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGRAC	SGST: General Switch Cost (reverse), absolute [mean(RB3TSMXROC, RB3TSMXRSC) – RB3TSMRC] corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCGRRC	SGST: General Switch Cost (reverse), relative (RB3TSCGRAC/ RB3TSMRC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING

RB3TSCLBAC	SGST: Local Switch Cost, absolute (RB3TSMXBSC-RB3TSMXBOC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLBRC	SGST: Local Switch Cost, relative (RB3TSCLBAC/RB3TSMXBOC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLNAC	SGST: Local Switch Cost (normal), absolute (RB3TSMXNSC-RB3TSMXNOC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLNRC	SGST: Local Switch Cost, (normal), relative (RB3TSCLNAC/RB3TSMXNOC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLRAC	SGST: Local Switch Cost (reverse), absolute (RB3TSMXRSC-RB3TSMXROC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
RB3TSCLRRC	SGST: Local Switch Cost,(reverse), relative (RB3TSCLRAC/RB3TSMXROC) corrected based on the metronome values	Latency (s) in seconds 98 = REFUSED OR MISSING
Filters and Phone Adjustments		
RB3TSFV	SGST: Filter out invalid cases	0 = NOT SELECTED 1 = SELECTED 8 = REFUSED OR MISSING
RB3TSFC	SGST: Filter out cases with low accuracy or extreme latencies	0=NOT SELECTED 1=SELECTED 8 = REFUSED OR MISSING
RB3TSM MM	Metronome correction (median of all available latencies)	Latency (s) in seconds 0 = PERFECT ACCURACY 98 = REFUSED OR MISSING

Montreal Cognitive Assessment (MoCA-BLIND)

A. Coding conventions for MoCA data

- a) PROBLEM HEARING/UNDERSTANDING/RECORDING(6's) 6, 96, 996, 9996
- b) DON'T KNOW (7's) 7, 97, 997, 9997
- c) REFUSED OR MISSING (8's) 8, 98, 998, 9998

B. Naming conventions for MoCA variables

- a) 1st letter – “R,” to indicate MIDUS Refresher
- b) 2nd letter – “B,” to indicate Wave 2
- c) 3rd character – “3,” indicates project number 3
- d) 4th letter indicates type of test: “M” = MoCA cognitive tests
- e) 5th, 6th, 7th, 8th, letters – indicates either:
 - i. The individual test:
 - IR: Immediate Recall
 - FDS: Forward Digit Span
 - BDS: Backward Digit Span
 - VIG: Vigilance
 - S7: Serial 7s
 - SREP: Sentence Repetition (also SR)
 - VBFL: Verbal Fluency
 - AB: Abstraction
 - DR: Delayed Recall
 - OR: Orientation (also wkdy, day, month, mth, year, yr)
 - ii. A Sub-Score: Variables with “TOT” as their last letters serve as the final score for the individual MoCA Tests.
 - iii. A Sequence: When a variable is numbered at the end, such as ‘1’, ‘2’, ‘3’, etc., the variable is part of a sequence and denotes its chronological position within that sequence.
 - iv. An Item Total: Variables with ‘TI’ as their last two characters represent total intrusions. Variables with ‘TR’ as their last two characters represent total intrusions. Variables with ‘TU’ as their last two characters represent total unique. (Specific to world list and verbal fluency). Variables with ‘ERR’ serve as the number of errors made in that task (specific to vigilance task).
 - iv. A flag variable: Variables with “FP” as their last letters serve as flag variables for potentially problematic cases. This variable indicates, by test, which cases were identified at Brandeis by our data cleaning as being problematic due to test disruption, interview equipment failures, or other problems. We recommend users exclude these tests for these specific cases.
 - vi. An accuracy variable: Variables ending in ‘C’ are accuracy variables and are on a 0-1 scale. A value of 1 represents a correct answer and a value of 0 represents and incorrect answer. The accuracy variables can be used to compute sub-scores.

Note: **Bold** variable names below indicate composite or total scale scores

VARIABLE NAME	VARIABLE LABEL	VALUES
MoCA Immediate Word Recall		
RB3MIRFP	MoCA Word Recall Immediate flagged problematic?	1=YES 2=NO 8=REFUSED OR MISSING
RB3MIR1T1 ... RB3MIR1T13	MoCA Immediate Word Recall; Trial 1; Response 1 (#1 ... #13, allows for max 5 correct responses + up to 8 intrusions/repetitions)	1 = WORD 1 2 = WORD 2 3 = WORD 3 4 = WORD 4 5 = WORD 5 80 = WORD FROM BTACT WORD LIST 90 = NON-LIST INTRUSION 96 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 98 = REFUSED OR MISSING
RB3MIRBTACT1	Total BTACT Words Recalled during MoCA Immediate Word Recall; Trial 1	Sum of all BTACT responses from RB3MIR1T1 to RB3MIR1T13 98 = REFUSED OR MISSING
RB3MIR1TR	Immediate Recall; Trial 1: Total Repetitions	Sum of all repeated responses from RB3MIR1T1 to RB3MIR1T13 98 = REFUSED OR MISSING
RB3MIR1TI	Immediate Recall; Trial 1: Total Intrusions	Sum of all non-list intrusions from RB3MIR1 to RB3MIR13 98 = REFUSED OR MISSING
RB3MIR2T1... RB3MIR2T13	MoCA Immediate Word Recall; Trial 2; Trial 2; Response 1 (#1 ... #13, allows for max 5 correct responses + up to 8 intrusions/repetitions)	1 = WORD 1 2 = WORD 2 3 = WORD 3 4 = WORD 4 5 = WORD 5 80 = WORD FROM BTACT WORD LIST 90 = NON-LIST INTRUSION 96 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 98 = REFUSED OR MISSING

RB3MIRBTACT2	Total BTACT Words Recalled during MoCA Immediate Word Recall; Trial 2	Sum of all BTACT responses from RB3MIR2T1 to RB3MIR2T13 98 = REFUSED OR MISSING
RB3MIR2TR	Immediate Recall; Trial 2: Total Repetitions	Sum of all repeated responses from RB3MIR2T1 to RB3MIR2T13 98 = REFUSED OR MISSING
RB3MIR2TI	Immediate Recall; Trial 2: Total Intrusions	Sum of all non-list intrusions from RB3MIR2T1 to RB3MIR2T13 98 = REFUSED OR MISSING
Digit Span		
RB3MDSFP	Digit Span Flagged problematic?	1 = YES 2 = NO 8 = REFUSED OR MISSING
RB3MFDS	Forward Digit Span	1 = 2-1-8-5-4 (CORRECT REPETITION) 2 = SOMETHING ELSE (INCORRECT REPETITION) 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 8 = REFUSED OR MISSING
RB3MBDS	Backward Digit Span	1= 2-4-7 (CORRECT REPETITION) 2= SOMETHING ELSE (INCORRECT REPETITION) 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 8 =REFUSED OR MISSING
RB3MFDSC	Forward Digit Span: Correct?	1 = YES, CORRECT
RB3MBDSC	Backward Digit Span: Correct?	2 = NO, INCORRECT 8 = REFUSED OR MISSING
RB3MDSTOT	Digit Span Total	Sum of correct responses for Digit Span 0 = NONE CORRECT 1 = ONE CORRECT 2 = TWO CORRECT 8 = REFUSED OR MISSING
Vigilance		
RB3MVIGFP	Vigilance Flagged Problematic?	1= YES 2 = NO 8 = REFUSED OR MISSING

RB3MVIGERR	Vigilance Number of Errors	Range: 0-29; possible error count 96 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 98 =REFUSED OR MISSING
RB3MVIGTOT	MoCA Vigilance Total Score: (possible minimum = 0, possible maximum = 1)	0 = TOTAL ERRORS GREATER THAN ONE 1 = TOTAL ERRORS LESS THAN OR EQUAL TO ONE 8 = REFUSED OR MISSING

Serial 7s

RB3MS7FP	Serial 7s Flagged Problematic?	1 = YES 2 = NO 8 = REFUSED OR MISSING
RB3MS71 RB3MS72 RB3MS73 RB3MS74 RB3MS75	Serial 7s; #1...#5	996 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 998 =REFUSED OR MISSING
RB3MS71C RB3MS72C RB3MS73C RB3MS74C RB3MS75C	Serial 7s #1...#5: Correct?	1 = YES, CORRECT 2 = NO, INCORRECT 8 =REFUSED OR MISSING
RB3MS7TOT	MoCA Serial 7s Total Score: (possible minimum = 0, maximum = 3)	0 = NO CORRECT SUBTRACTIONS 1 = ONE CORRECT SUBTRACTION 2 = TWO OR THREE CORRECT SUBTRACTIONS 3 = FOUR OR FIVE CORRECT SUBTRACTIONS 8 =REFUSED OR MISSING

Sentence Repetition

RB3MSREPPF	Sentence Repetition Flagged Problematic?	1 = YES 2 = NO 8 = REFUSED OR MISSING
RB3MSREP1 RB3MSREP2	Sentence Repetition; #1 ... #2	1 = EXACT REPETITION 2 = SOMETHING ELSE 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 8 = REFUSED OR MISSING

RB3MSREP1C RB3MSREP2C	Sentence Repetition; #1 ... #2: Correct?	1 = YES, CORRECT 2 = NO, INCORRECT 8 = REFUSED OR MISSING
RB3MSREPTOT	MoCA Sentence Repetition Total Score	Sum of correct repetitions 0 = NO CORRECT REPETITIONS 1 = ONE CORRECT REPETITION 2 = TWO CORRECT REPETITIONS 8 = REFUSED OR MISSING

Verbal Fluency

RB3MVBFFP	Verbal Fluency Flagged Problematic?	1 = YES 2 = NO 8 = REFUSED OR MISSING
RB3MVBFL1... RB3MVBFL40	Verbal Fluency; Response 1....40	0 = INVALID WORD (INTRUSION) 1 = VALID WORD 2 = REPETITION 8 = REFUSED OR MISSING
RB3MVBFTU	Verbal Fluency: Total Unique	Sum of valid F words from RB3MVBFL1 through RB3MVBFL40 98 = REFUSED OR MISSING
RB3MVBFTI	Verbal Fluency: Total Intrusions	Sum of non-list intrusions from RB3MVBFL1 through RB3MVBFL40 98 = REFUSED OR MISSING
RB3MVBFTTR	Verbal Fluency: Total Repetitions	Sum of repetitions from RB3MVBFL1 through RB3MVBFL40 98 = REFUSED OR MISSING
RB3MVBFTOT	Verbal Fluency: Total Score	1 = EQUAL TO OR GREATER THAN 11 VALID WORDS 0 = LESS THAN 11 VALID WORDS 8 = REFUSED OR MISSING

Abstraction

RB3MABSFP	Abstraction flagged problematic?	1 = YES 2 = NO 8 = REFUSED OR MISSING
RB3MAB1 RB3MAB2	Abstraction 1 Abstraction 2	1 = CORRECT CATEGORY 2 = INCORRECT CATEGORY 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 8 = REFUSED OR MISSING
RB3MAB1C RB3MAB2C	Abstraction 1...2: Correct?	1 = YES, CORRECT 2 = NO, CORRECT 8 = REFUSED OR MISSING

RB3MABTOT	MoCA Abstraction Total Score	0 = NONE CORRECT 1 = ONE CORRECT 2 = TWO CORRECT 8 = REFUSED OR MISSING
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Delayed Word Recall

RB3MDRFP	Delayed Recall Flagged Problematic?	1 = YES 2 = NO 8 = REFUSED OR MISSING
RB3MDR1...	MoCA Delayed Word	1 = WORD 1
RB3MDR13	Recall; Response 1 ... 13 (allows for max 5 correct responses + up to 8 Intrusions/repetitions)	2 = WORD 2 3 = WORD 3 4 = WORD 4 5 = WORD 5 80 = WORD FROM BTACT WORD LIST 90 = NON-LIST INTRUSION 96 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 98 = REFUSED OR MISSING
RB3MDRTR	MoCA Delayed Recall: Total repetitions	Sum of repetitions from RB3MDR1 through RB3MDR13 98 = REFUSED OR MISSING
RB3MDRTI	MoCA Delayed Recall: Total Intrusions	Sum of intrusions from RB3MDR1 through RB3MDR13 98 = REFUSED OR MISSING
RB3MDRTOT	MoCA Delayed Recall: Total score	Sum of unique words from RB3MDR1 through RB3MDR13 98 = REFUSED OR MISSING

Delayed Recall: Memory Index Score

RB3MDRHINTF	MoCA Delayed Recall; Word 1 hint response	1 = WORD 1 2 = DOES NOT SAY WORD 1 8 = REFUSED OR MISSING
RB3MDRHINTV	MoCA Delayed Recall; Word 2 hinted response	1 = WORD 2 2 = DOES NOT SAY WORD 2 8 = REFUSED OR MISSING
RB3MDRHINTC	MoCA Delayed Recall; Word 3 hinted response	1 = WORD 3 2 = DOES NOT SAY WORD 3 8 = REFUSED OR MISSING
RB3MDRHINTD	MoCA Delayed Recall; Word 4 hinted response	1 = WORD 4 2 = DOES NOT SAY WORD 4 8 = REFUSED OR MISSING

RB3MDRHINTR	MoCA Delayed Recall; Word 5 hinted response	1 = WORD 5 2 = DOES NOT SAY WORD 5 8 = REFUSED OR MISSING
RB3MDRMCF	MoCA Delayed Recall; Word 1 multiple choice response	1 = WORD 1 2 = DOES NOT SAY WORD 1 8 = REFUSED OR MISSING
RB3MDRMCV	MoCA Delayed Recall; Word 2 multiple choice response	1 = WORD 2 2 = DOES NOT SAY WORD 2 8 = REFUSED OR MISSING
RB3MDRMCC	MoCA Delayed Recall; Word 3 multiple choice response	1 = WORD 3 2 = DOES NOT SAY WORD 3 8 = REFUSED OR MISSING
RB3MDRMCD	MoCA Delayed Recall; Word 4 multiple choice response	1 = WORD 4 2 = DOES NOT SAY WORD 4 8 = REFUSED OR MISSING
RB3MDRMCR	MoCA Delayed Recall; Word 5 multiple choice	1 = WORD 5 2 = DOES NOT SAY WORD 5 8 = REFUSED OR MISSING
RB3MDRMIS	MoCA Delayed Recall Memory Index Total Score	Range = 0 – 15 98 = REFUSED OR MISSING

Orientation

RB3MORFP	Orientation flagged problematic?	1 = YES 2 = NO 8 = REFUSED OR MISSING
RB3MWKDY	Orientation; Weekday	1 = MONDAY 2 = TUESDAY 3 = WEDNESDAY 4 = THURSDAY 5 = FRIDAY 6 = SATURDAY 7 = SUNDAY 96 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 97 = DON'T KNOW 98 = REFUSED OR MISSING
RB3MDATE	Orientation; Date	96 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 97 = DON'T KNOW 98 = REFUSED OR MISSING

RB3MMTH	Orientation; Month	1 = JANUARY 2 = FEBRUARY 3 = MARCH 4 = APRIL 5 = MAY 6 = JUNE 7 = JULY 8 = AUGUST 9 = SEPTEMBER 10 = OCTOBER 11 = NOVEMBER 12 = DECEMBER 96 = PROBLEM HEARING/UNDERSTANDING/RECORDING 97 = DON'T KNOW 98 = REFUSED OR MISSING
RB3MYEAR	Orientation; Year	9996 = PROBLEM HEARING/UNDERSTANDING/RECORDING 9997 = DON'T KNOW 9998 = REFUSED OR MISSING
RB3MCITY	Orientation; Current City	1 = ANSWERS WITH A CITY 2 = ANSWERS WITH SOMETHING ELSE 6 = PROBLEM HEARING/UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING
RB3MSTATE	Orientation; Current State	1 = ANSWERS WITH A STATE 2 = ANSWERS WITH SOMETHING ELSE 6 = PROBLEM HEARING/UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING
RB3MWKDYC	Orientation; Weekday: Correct?	0 = NO, INCORRECT 1 = YES, CORRECT 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING
RB3MDATEC	Orientation; Date: Correct?	0 = NO, INCORRECT 1 = YES, CORRECT 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING

RB3MMTHC	Orientation; Month: Correct?	0 = NO, INCORRECT 1 = YES, CORRECT 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING
RB3BYEARC	Orientation; Year:	0 = NO, INCORRECT 1 = YES, CORRECT 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING
RB3MCITYC	Orientation; City: Correct?	0 = NO, INCORRECT 1 = YES, CORRECT 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING
RB3MSTATEC	Orientation; State: Correct?	0 = NO, INCORRECT 1 = YES, CORRECT 6 = PROBLEM HEARING/ UNDERSTANDING/RECORDING 7 = DON'T KNOW 8 = REFUSED OR MISSING
RB3MORTOT	MoCA Orientation Total Score	1 = 1 CORRECT 2 = 2 CORRECT 3 = 3 CORRECT 4 = 4 CORRECT 5 = 5 CORRECT 6 = 6 CORRECT 8 = REFUSED OR MISSING
MoCA Total Final Score		
RB3MOCATOT	MoCA Total score	Range: 0-22 98 = REFUSED OR MISSING
RB3MOCATOTADJ	MoCA Final score with education adjustment	Range: 0-22 98 = REFUSED OR MISSING
RB3MOCAPARTIAL	MoCA Partial total score	Range: 0-21 98 = REFUSED OR MISSING